

NEET (UG) - 2026: MASTER ANSWER KEY & EXPLANATIONS

Test Booklet Code: PT-2 | 180 Questions with Detailed NCERT Reference

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DETAILED STEP-BY-STEP EXPLANATIONS

Question 1 [Answer: (2)] - Physics (NCERT Page 80, 81)

Explanation: Work done by force F is $W = F^2/(2k)$. Since $W \propto 1/k$ and $k_P = 2k_Q$, $W_P/W_Q = k_Q/k_P = 1/2 \Rightarrow W_Q = 2W_P$.

Question 2 [Answer: (2)] - Physics (NCERT Page 78)

Explanation: $F = -\nabla U = -[(\partial U/\partial x)\hat{i} + (\partial U/\partial y)\hat{j} + (\partial U/\partial z)\hat{k}]$. At $(1,1,1)$: $\partial U/\partial x = 6xy = 6$; $\partial U/\partial y = 3x^2 - 4z = -1$; $\partial U/\partial z = 4z - 3x^2 = 1$.

Question 3 [Answer: (1)] - Physics (NCERT Page 133, 134)

Explanation: At equator, $g_{\text{eff}} = g - R\omega^2 = 0 \Rightarrow \omega = \sqrt{g/R} = \sqrt{(10/6.4 \times 10^6)} = 1/(800) = 1.25 \times 10^{-3} \text{ rad/s}$.

Question 4 [Answer: (1)] - Physics (NCERT Page 75)

Explanation: $W = \int_{\infty}^x (kx + 2x^3) dx = [kx^2/2 + 2x^4/4]_{\infty}^x = (1/2)kL^2 + (1/4)^2Lt$.

Question 5 [Answer: (1)] - Physics (NCERT Page 130)

Explanation: Net gravitational force towards circumcentre: $F_{\text{net}} = 2(Gm^2/a^2) \cos 30^\circ = \sqrt{3} Gm^2/a^2$. Radius $r = a/\sqrt{3}$.

Question 6 [Answer: (2)] - Physics (NCERT Page 130)

Explanation: Orbital radius $r^* = R + R = 2R$. Orbital radius $r_c = R + 7R = 8R$. By Kepler's 3rd law, $T \propto r^{3/2}$. $T_c/T^* = (8R/2R)^{3/2} = 8$.

Question 7 [Answer: (2)] - Physics (NCERT Page 83)

Explanation: $P = Fv = (m \frac{dv}{ds})v = m v^2 (\frac{dv}{ds})$. Thus $v^2 dv = (P/m) ds$. Integrating: $v^3/3 = \int P ds \Rightarrow v = \sqrt[3]{3 \int P ds}$.

Question 8 [Answer: (1)] - Physics (NCERT Page 134)

Explanation: From conservation of mechanical energy: Loss in gravitational PE = Gain in elastic PE $\Rightarrow mg(h+x) = \frac{1}{2}kx^2$.

Question 9 [Answer: (1)] - Physics (NCERT Page 136)

Explanation: $v_e = \sqrt{2GM/R} \propto R^{-1/2}$. $v_e/v_e = -1/2 (\Delta R/R) = -1/2 (-1\%) = +0.5\%$.

Question 10 [Answer: (1)] - Physics (NCERT Page 84, 85)

Explanation: Final velocity of mass m : $v^* = [(m-3m)/(m+3m)]u = -u/2$. Initial KE = $(1/2)mu^2$. Final KE = $(1/2)m(u^2/4) = (1/8)mu^2$.

Question 11 [Answer: (2)] - Physics (NCERT Page 130)

Explanation: Mass enclosed $M(r) = \int_0^r 4\pi r^2 \rho(r) dr = 4\pi \int_0^r r^2 \rho(r) dr = 4\pi \int_0^r r^2 \rho_0 (1 - \alpha r) dr$. Gravitational field $E(r) = GM(r)/r^2$.

Question 12 [Answer: (1)] - Physics (NCERT Page 119)

Explanation: By conservation of angular momentum: $L = I\omega = (2I\omega_f) \Rightarrow \omega_f = \omega/2$. Final KE = $L^2 / (2I_f) = L^2 / (4I)$.

Question 13 [Answer: (1)] - Physics (NCERT Page 134, 135)

Explanation: $E = -dV/dx = -d/dx[-C(x^2 + a^2)^{-1/2}] = C(-1/2)(x^2 + a^2)^{-3/2}(2x) = -Cx / (x^2 + a^2)^{3/2}$. Magnitude = $Cx / (x^2 + a^2)^{3/2}$.

Question 14 [Answer: (3)] - Physics (NCERT Page 78)

Explanation: At bottom B: $v^2 = 2gR$. Normal force $N - mg = mv^2/R = 2mg \Rightarrow N = 3mg$ directed radially inwards towards centre.

Question 15 [Answer: (1)] - Physics (NCERT Page 130)

Explanation: $dF = G m dm / x^2 = G m (M/L dx) / x^2$. $F = (GMm/L) \int_{d+L}^d x^{-2} dx = (GMm/L) [1/d - 1/(d+L)] = GMm/L \cdot L/(d(d+L))$.

Question 16 [Answer: (1)] - Physics (NCERT Page 83)

Explanation: $v = kx \Rightarrow a = v (dv/dx) = (kx)(k/(2x)) = k^2/2$. Force $F = ma = mk^2/2$. Power $P = Fv = (mk^2/2)(kx) = mk^3x/2$.

Question 17 [Answer: (1)] - Physics (NCERT Page 106)

Explanation: $\vec{A} \times \vec{r} \times \vec{F} = \text{determinant} [\hat{i}, 5, k; 1, -2, 1; 2, 3, -4] = \hat{i}(8-3) - 5(-4-2) + k(3-(-4)) = 5\hat{i} + 65\hat{j} + 7k\hat{k}$.

Question 18 [Answer: (1)] - Physics (NCERT Page 115, 116)

Explanation: Side of equilateral triangle $d = 2R$. Distance of each centre from centroid $r = d/\sqrt{3} = 2R/\sqrt{3}$. For each sphere: $(26/15)MR^2$. For 3 spheres: $I_{\text{total}} = 3 \times (26/15)MR^2 = (26/5)MR^2$.

Question 19 [Answer: (1)] - Physics (NCERT Page 130)

Explanation: Distance of each vertex from centre $r = a/\sqrt{2}$. Total potential $V = 4 \times (-GM/r) = 4 \times (-GM/(a/\sqrt{2})) = -4\sqrt{2}GM/a$.

Question 20 [Answer: (1)] - Physics (NCERT Page 76, 78)

Explanation: By Work-Energy theorem: $W_{\text{total}} = \Delta K \Rightarrow W_g + W_{\text{res}} = K_f - 0 \Rightarrow mgh - fh = K_f$.

Question 21 [Answer: (1)] - Physics (NCERT Page 96)

Explanation: $X_{\text{cm}} = \sum (m_i x_i) / \sum m_i = [m \cdot d(1^2 + 2^2 + \dots + n^2)] / [m(1 + 2 + \dots + n)] = [n(n+1)(2n+1)/6] / [n(n+1)/2] = (2n+1)/3$.

Question 22 [Answer: (1)] - Physics (NCERT Page 137)

Explanation: Orbital velocity $v_{\text{orb}} = \sqrt{GM/r} \propto 1/\sqrt{r}$. Since $r_A = 4r_B$, $v_B/v_A = \sqrt{(r_A/r_B)} = \sqrt{4} = 2 \Rightarrow v_B = 2v_A$.

Question 23 [Answer: (1)] - Physics (NCERT Page 106)

Explanation: $K = (1/2)mv^2 = c s^2 \Rightarrow v = \sqrt{2cs}$. $dv/ds = \text{const}$. Tangential acceleration $a_t = v (dv/ds) = s$.

Question 24 [Answer: (1)] - Physics (NCERT Page 120)

Explanation: For rolling down an incline: $a = g \sin \theta / (1 + I/MR^2)$. For solid cylinder, $I/MR^2 = 1/2$. So $a = g \sin \theta / (1 + 1/2) = (2/3)g \sin \theta$.

Question 25 [Answer: (1)] - Physics (NCERT Page 76)

Explanation: $v(t) = dx/dt = 6t^2 - 6t$. At $t = 0$: $v(0) = 0$. At $t = 2$ s: $v(2) = 6(4) - 6(2) = 12$ m/s. Work done = $\Delta K = (1/2)mv^2 = (1/2)(1)(12)^2 = 72$ J.

Question 26 [Answer: (1)] - Physics (NCERT Page 74)

Explanation: $mv^2/r = k / r^2 \Rightarrow v^2 = k / (m r^3)$. Since $v = 2\pi r/T \Rightarrow 4\pi^2 r^2/T^2 = k / (m r^3) \Rightarrow T^2 \propto r^{-1}$.

Question 27 [Answer: (1)] - Physics (NCERT Page 109)

Explanation: The center of mass of the stick is at the 50 cm mark. Distance from pivot (40 cm): Stick CM is at $50 - 40 = 10$ cm to the right. Mass m is at $40 - 10 = 30$ cm to the left. Balancing torques: $m \times 30 = 150 \times 10 \Rightarrow m = 50$ g.

Question 28 [Answer: (1)] - Physics (NCERT Page 107)

Explanation: $\ddot{L} = dL/dt \Rightarrow L = \ddot{L} \times t = 20 \text{ N}\cdot\text{m} \times 5 \text{ s} = 100 \text{ kg}\cdot\text{m}^2/\text{s}$.

Question 29 [Answer: (1)] - Physics (NCERT Page 138)

Explanation: Distance to any element of the ring is $r = \sqrt{(R^2 + x^2)} = \sqrt{(R^2 + 3R^2)} = 2R$. Potential $V = -GM / r = -GM / (2R)$.

Question 30 [Answer: (1)] - Physics (NCERT Page 117)

Explanation: $\theta(2) = (1/2)\alpha(2)^2 = 2\pi = 20 \Rightarrow \alpha = 10 \text{ rad/s}^2$. Angle in 4 seconds: $\theta(4) = (1/2)(10)(4)^2 = 80 \text{ rad}$. Angle in radians is 80 .

Question 31 [Answer: (1)] - Physics (NCERT Page 130)

Explanation: Initial PE $U_i = -3Gm^2/L$. Final PE $U_f = -3Gm^2/(2L)$. Work done $W = U_f - U_i = -3Gm^2/(2L) - (-3Gm^2/L) = 3Gm^2/(2L)$.

Question 32 [Answer: (1)] - Physics (NCERT Page 84, 85)

Explanation: Initial KE = $(1/2)mv^2$. Common velocity after collision = $v/5$. Final KE = $(1/2)(5m)(v/5)^2 = (1/5) KE_{\text{initial}}$.

Question 33 [Answer: (1)] - Physics (NCERT Page 106, 107)

Explanation: $r \times \hat{i} = x \hat{i} + b \hat{j}$. $p \times = m v \hat{i}$. $L \times = r \times \times p \times = (x \hat{i} + b \hat{j}) \times (mv \hat{i}) = -mbv \hat{k}$. This is independent of x , hence constant.

Question 34 [Answer: (1)] - Physics (NCERT Page 129)

Explanation: $g = (4/3) \pi G \rho R \Rightarrow g \propto R$. $g_A / g_B = (\rho_A / \rho_B) \times (R_A / R_B) = (1/2) \times (2/1) = 1 : 1$.

Question 35 [Answer: (1)] - Physics (NCERT Page 84)

Explanation: $v_{\text{comb}} = (0.02 \times 400) / (0.02 + 1.98) = 8 / 2 = 4$ m/s. Height $h = v^2 / (2g) = 16 / 20 = 0.8$ m. Cord length = 0.2 m. Wait, if $h = 0.5$ m, $\theta = 60^\circ$. If $h = 0.8$ m, $\cos \theta = 0.2$ (approx 78°).

Question 36 [Answer: (1)] - Physics (NCERT Page 83)

Explanation: Total input power $P_{\text{in}} = (dm/dt) g h = 15 \times 10 \times 50 = 7500 \text{ W} = 7.5 \text{ kW}$. Efficiency $\eta = 90\%$. Useful output power $P_{\text{out}} = 0.9 \times 7500 = 6750 \text{ W}$.

Question 37 [Answer: (1)] - Physics (NCERT Page 128)

Explanation: By conservation of angular momentum about the Sun: $L = m v \times r = m v_r r \Rightarrow v \propto 1/r$.

Question 38 [Answer: (1)] - Physics (NCERT Page 120)

Explanation: Acceleration $a = g \sin \theta / (1 + k^2/R^2)$. For solid sphere, $k^2/R^2 = 2/5 = 0.4$. For hollow cylinder, $k^2/R^2 = 1$. Linear acceleration is larger and it reaches the bottom first.

Question 39 [Answer: (1)] - Physics (NCERT Page 78)

Explanation: $V(2) = 16 - 8 = 8$ J. Total mechanical energy $E = 8$ J. Speed is maximum where $V(x)$ is minimum: $dV/dx = 0 \Rightarrow V_{\text{min}} = 8 - (-1) = 9$ J. $(1/2)mv^2 = 9 \Rightarrow v^2 = 18 \Rightarrow v = \sqrt{18} = 3\sqrt{2}$ m/s.

Question 40 [Answer: (1)] - Physics (NCERT Page 76)

Explanation: $W = \int_{10}^{10+0.5} x dx = [10x + 0.25x^2]_{10}^{10.5} = 20 + 0.25(4) = 21$ J.

Question 41 [Answer: (1)] - Physics (NCERT Page 114, 115)

Explanation: Moment of inertia about diameter is $(1/4)MR^2$. By parallel axes theorem, about a tangent in plane: $I = (1/4)MR^2 + MR^2 = (5/4)MR^2$. Radius of gyration $k = \sqrt{(I/M)} = (\sqrt{5}/2) R$.

Question 42 [Answer: (1)] - Physics (NCERT Page 119, 122)

Explanation: Conservation of angular momentum: $L = I \cdot \dot{\theta} + (2I \cdot)(2\dot{\theta}) = 5I \cdot \dot{\theta}$. Total moment of inertia $I = I_1 + 2I_2$.

Question 43 [Answer: (1)] - Physics (NCERT Page 136, 137)

Explanation: $v_{\text{inf}}^2 = v^2 - v_e^2 = (2v_e)^2 - v_e^2 = 3v_e^2 \Rightarrow v_{\text{inf}} = v_e \sqrt{3} = 11.2 \sqrt{3} \text{ km/s}$.

Question 44 [Answer: (1)] - Physics (NCERT Page 130)

Explanation: Mass of original sphere = M . Mass of cavity $m_c = M(r/R)^3 = M(1/2)^3 = M/8$. Distance of cavity centre $x_c = R/2$. Mass of remaining portion $m_r = 7M/8$. $x_{\text{cm}} = -(m_c x_c) / m_r = -(M/8 \times R/2) / (7M/8) = -R/14$.

Question 45 [Answer: (1)] - Physics (NCERT Page 74)

Explanation: Vertical height $h = d \sin 30^\circ = 5 \times 0.5 = 2.5$ m. Work done against gravity = $m g h = 2 \times 10 \times 2.5 = 50$ J.

Question 46 [Answer: (1)] - Chemistry (NCERT Page 143)

Explanation: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$. $\Delta H^\circ = [\Delta H^\circ(\text{CO}_2) + 2\Delta H^\circ(\text{H}_2\text{O})] - [\Delta H^\circ(\text{CH}_4)] = [-393.5 + 2(-285.8)] - [-74.6] = -890.3 \text{ kJ/mol}$.

Question 47 [Answer: (1)] - Chemistry (NCERT Page 120)

Explanation: SF_6 has sp^3d^2 hybridization with regular octahedral geometry where all 6 S-F bond lengths are equivalent.

Question 48 [Answer: (1)] - Chemistry (NCERT Page 153)

Explanation: Bonds broken: 1 C=C (614) + 1 H-H (436) = 1050 kJ. Bonds formed: 1 C-C (347) + 2 C-H (2×414) = 875 kJ.

Question 49 [Answer: (4)] - Chemistry (NCERT Page 112)

Explanation: Higher positive charge and smaller cation size increase polarizing power. SiCl_4 has highest charge and SiCl_4 .

Question 50 [Answer: (1)] - Chemistry (NCERT Page 142)

Explanation: $w_{\text{rev}} = -2.303 n R T \log(V_2/V_1) = -2.303 \times 2 \times 8.314 \times 300 \times \log(10) = -11488 \text{ J} = -11.49 \text{ kJ}$.

Question 51 [Answer: (1)] - Chemistry (NCERT Page 125)

Explanation: ClF_3 : 3 bp + 2 lp = T-shaped; XeF_4 : 4 bp + 2 lp = Square planar; SF_6 : 6 bp + 0 lp = Octahedral; BF_3 : 3 bp + 0 lp = Trigonal planar.

Question 52 [Answer: (2)] - Chemistry (NCERT Page 111)

Explanation: CO (14 electrons) has bond order 3. NO^+ (14 electrons) also has bond order 3.

Question 53 [Answer: (1)] - Chemistry (NCERT Page 131)

Explanation: Both statements are true facts: each H-F hydrogen bond is individually stronger than an H-O bond due to higher electronegativity of F. Because of 4 H-bonds per molecule forming an extensive 3D network.

Question 54 [Answer: (1)] - Chemistry (NCERT Page 143)

Explanation: $n_g = 2 - (1 + 3) = -2$. $\Delta H = \Delta U + n_g R T$ $\Delta U = \Delta H - n_g R T = -92.4 - [(-2) \times 8.314 \times 298 / 1000] = -72.4 \text{ kJ mol}^{-1}$.

Question 55 [Answer: (1)] - Chemistry (NCERT Page 143)

Explanation: $\Delta H = \Delta U + n_g R T$. When $n_g = 0$ (no net change in moles of gaseous species), $\Delta H = \Delta U$.

Question 56 [Answer: (1)] - Chemistry (NCERT Page 114)

Explanation: Standard VSEPR postulate: $\text{lp-lp} > \text{lp-bp} > \text{bp-bp}$ repulsion.

Question 57 [Answer: (1)] - Chemistry (NCERT Page 162)

Explanation: $\Delta S_{\text{fusion}} = \Delta H_{\text{fusion}} / T = 6000 \text{ J mol}^{-1} / 273.15 \text{ K} = 21.98 \text{ J K}^{-1} \text{ mol}^{-1}$.

Question 58 [Answer: (1)] - Chemistry (NCERT Page 111)

Explanation: BeCl_2 is linear (dipoles cancel) and BCl_3 is trigonal planar (dipoles cancel), so both have zero dipole moment.

Question 59 [Answer: (1)] - Chemistry (NCERT Page 102)

Explanation: In Lewis structure of OF_2 : central O has 1 lone pair (2 electrons) and 2 bonds (4 shared electrons). For F, 3 lone pairs (6 electrons) and 1 bond (2 shared electrons).

Question 60 [Answer: (1)] - Chemistry (NCERT Page 162)

Explanation: Second Law of Thermodynamics states that spontaneous processes in isolated systems increase entropy, and reversible cyclic processes leave the universe entropy unchanged ($\Delta S_{\text{univ}} = 0$).

Question 61 [Answer: (1)] - Chemistry (NCERT Page 120)

Explanation: IF_5 : 5 bp + 1 lp = steric number 6 (sp^3d^2); PCl_5 : 5 bp + 0 lp = steric number 5 (sp^3d); XeOF_4 : 5 bp + 1 lp = steric number 6 (sp^3d^2).

Question 62 [Answer: (1)] - Chemistry (NCERT Page 154)

Explanation: $\Delta H^\circ = 2\Delta_f H^\circ(\text{SO}_2) - [2\Delta_f H^\circ(\text{SO}_3) + 0]$ $-198 = 2(-395.7) - 2\Delta_f H^\circ(\text{SO}_3)$ $2\Delta_f H^\circ(\text{SO}_3) = -791.4 + 198 = -593.4$ $\Delta_f H^\circ(\text{SO}_3) = -296.7 \text{ kJ mol}^{-1}$.

Question 63 [Answer: (1)] - Chemistry (NCERT Page 125)

Explanation: O_3 (15 e⁻) = 2.5; O_2 (16 e⁻) = 2.0; O_2^+ (15 e⁻) = 1.5; O_2^{2+} (14 e⁻) = 1.0.

Question 64 [Answer: (1)] - Chemistry (NCERT Page 154)

Explanation: In an adiabatic process $q = 0$. In expansion $w < 0$ (gas does work on surroundings), so $\Delta U = w < 0$, which is correct.

Question 65 [Answer: (1)] - Chemistry (NCERT Page 154)

Explanation: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$, ($\Delta H = -393.5$); $\text{CO} + (1/2)\text{O}_2 \rightarrow \text{CO}_2$, ($\Delta H = -283.0$). Subtracting eq 2 from eq 1 gives $\text{C} + (1/2)\text{O}_2 \rightarrow \text{CO}$, ($\Delta H = -110.5$).

Question 66 [Answer: (1)] - Chemistry (NCERT Page 154)

Explanation: $\Delta G = \Delta H - T\Delta S$. At equilibrium $\Delta G = 0$ $T = \Delta H / \Delta S = -40000 \text{ J} / (-50 \text{ J K}^{-1}) = 800 \text{ K}$. Since ΔS is negative, the process is non-spontaneous.

Question 67 [Answer: (1)] - Chemistry (NCERT Page 143)

Explanation: Isochoric: $w = 0$ $q_v = \Delta U$; Isobaric: $q_p = \Delta H$; Isothermal: $\Delta T = 0$ $\Delta U = 0$; Adiabatic: $q = 0$.

Question 68 [Answer: (1)] - Chemistry (NCERT Page 142)

Explanation: $n_g = 2 - 1 = 1$. $\Delta U^\circ = \Delta H^\circ - n_g R T = 124.0 - (1 \times 8.314 \times 298 / 1000) = 124.0 - 2.48 = 121.52 \text{ kJ mol}^{-1}$.

Question 69 [Answer: (1)] - Chemistry (NCERT Page 120)

Explanation: SF_6 has 6 bp + 0 lp = 6 (sp^3d^2). XeF_4 has 4 bp + 2 lp = 6 (sp^3d^2).

Question 70 [Answer: (1)] - Chemistry (NCERT Page 144)

Explanation: Density, temperature, refractive index, and molar heat capacity are all independent of the size/quantity of the system, hence intensive.

Question 71 [Answer: (1)] - Chemistry (NCERT Page 158)

Explanation: $\Delta G = \Delta H - T\Delta S$. When $\Delta H > 0$ and $\Delta S > 0$, at high temperatures $T\Delta S > \Delta H$, making $\Delta G < 0$ (spontaneous).

Question 72 [Answer: (1)] - Chemistry (NCERT Page 157)

Explanation: At constant T and P, a process is spontaneous if and only if the change in Gibbs free energy of the system is negative.

Question 73 [Answer: (1)] - Chemistry (NCERT Page 125)

Explanation: Configuration for O, (16 e⁻): $1s^2 2s^2 2p^4$ ($2p_x^2 = 2p_y^2$) ($2p_x^1 = 2p_y^1$). Antiparallel (from Δm_l) = 6.

Question 74 [Answer: (1)] - Chemistry (NCERT Page 125)

Explanation: C, (12 electrons): $1s^2 2s^2 2p^2$ ($2p_x^2 = 2p_y^2$). All electrons are paired (diamagnetic) and diamagnetic.

Question 75 [Answer: (1)] - Chemistry (NCERT Page 125)

Explanation: N, has 14 electrons; all electrons are completely paired in bonding and antibonding MOs, hence it is diamagnetic.

Question 76 [Answer: (1)] - Chemistry (NCERT Page 112)

Explanation: Lattice energy $\propto (q_1 \cdot q_2) / (r_1 + r_2)$. Li⁺ and F⁻ have the smallest ionic radii, giving LiF the maximum lattice energy.

Question 77 [Answer: (1)] - Chemistry (NCERT Page 112)

Explanation: Both statements are correct. In both H₂O and NH₃, the central atom is sp³ hybridized. H₂O has 2 lone pairs, which compresses the bond angle in H₂O to 104.5°.

Question 78 [Answer: (1)] - Chemistry (NCERT Page 120)

Explanation: BrF₅ has 5 bond pairs and 1 lone pair (steric number 6, sp³d²), which gives a square pyramidal geometry.

Question 79 [Answer: (1)] - Chemistry (NCERT Page 111)

Explanation: o-Nitrophenol forms intramolecular H-bonding (chelation), whereas p-nitrophenol forms intermolecular H-bonding.

Question 80 [Answer: (1)] - Chemistry (NCERT Page 107, 112)

Explanation: In CCl₄, the 4 identical tetrahedral bond dipoles sum to zero vectorially, giving $\mu = 0$. In CHCl₃ the dipoles do not cancel.

Question 81 [Answer: (1)] - Chemistry (NCERT Page 158)

Explanation: For spontaneity $\Delta G = \Delta H - T\Delta S < 0 \Rightarrow T > \Delta H / \Delta S = 30000 \text{ J} / (100 \text{ J K}^{-1}) = 300 \text{ K}$.

Question 82 [Answer: (1)] - Chemistry (NCERT Page 112)

Explanation: Cs has the lowest electronegativity (0.7) and F has the highest (4.0). The maximum difference in electronegativity characterizes ionic character.

Question 83 [Answer: (1)] - Chemistry (NCERT Page 143)

Explanation: $w = -P_{\text{ext}} \Delta V = -2.0 \text{ atm} \times (3.0 - 1.0 \text{ L}) = -4.0 \text{ L} \cdot \text{atm} = -4.0 \times 101.3 \text{ J} = -405.2 \text{ J}$. $\Delta U = q + w = +250 \text{ J} - 405.2 \text{ J} = -155.2 \text{ J}$.

Question 84 [Answer: (1)] - Chemistry (NCERT Page 154)

Explanation: Equation (1) - Equation (2): $C(\text{graphite}) - C(\text{diamond}) = -393.5 - (-395.4) = +1.9 \text{ kJ mol}^{-1}$ for C(graphite).

Question 85 [Answer: (1)] - Chemistry (NCERT Page 125)

Explanation: O₂ has 15 electrons, bond order = $(10 - 5)/2 = 2.5$ (fractional).

Question 86 [Answer: (1)] - Chemistry (NCERT Page 105)

Explanation: Both statements are true. Odd-electron molecules and hypervalent molecules with expanded octets (PCl₅) are exceptions to the octet rule.

Question 87 [Answer: (1)] - Chemistry (NCERT Page 131)

Explanation: All three statements A, B, and C are factual and accurate according to NCERT.

Question 88 [Answer: (1)] - Chemistry (NCERT Page 120)

Explanation: CH₄ is sp³, C₂H₄ (ethene) is sp², C₂H₂ (ethyne) is sp, and PCl₅ is sp³d.

Question 89 [Answer: (1)] - Chemistry (NCERT Page 114, 163)

Explanation: Heat and work are path functions (not state functions). A, C, and D are all correct.

Question 90 [Answer: (1)] - Chemistry (NCERT Page 125)

Explanation: $\Delta G^\circ = -RT \ln K$; $\Delta H = \Delta U + P\Delta V$; $\Delta S = q_{\text{rev}} / T$; $\Delta G = \Delta H - T\Delta S$.

Question 91 [Answer: (1)] - Biology (NCERT Page 95)

Explanation: RER has ribosomes studded on its outer surface and is actively involved in protein synthesis and secretion.

Question 92 [Answer: (1)] - Biology (NCERT Page 60)

Explanation: Monocot leaves typically exhibit parallel venation, whereas reticulate venation is characteristic of dicotyledonous leaves.

Question 93 [Answer: (1)] - Biology (NCERT Page 89)

Explanation: Penicillin targets peptidoglycan cell wall biosynthesis. Since mycoplasmas completely lack a cell wall, they are naturally unaffected by penicillin.

Question 94 [Answer: (1)] - Biology (NCERT Page 66)

Explanation: The seed coat of a dicot seed has two layers: the outer thick layer is the testa and the inner thin layer is the tegmen.

Question 95 [Answer: (1)] - Biology (NCERT Page 65)

Explanation: In marginal placentation, the placenta forms a ridge along the ventral suture of the ovary and ovules are borne along this ridge, as in pea.

Question 96 [Answer: (1)] - Biology (NCERT Page 71, 77)

Explanation: Bulliform cells roll leaves in grasses to reduce water loss; Trichomes prevent water loss on stem; Casparian strips are in endodermis; Lenticels facilitate gas exchange in woody stems.

Question 97 [Answer: (1)] - Biology (NCERT Page 96, 97)

Explanation: Both statements are verbatim true as per NCERT Cell: Unit of Life chapter.

Question 98 [Answer: (1)] - Biology (NCERT Page 66)

Explanation: Mango and coconut are drupes, developing from monocarpellary superior ovaries and are one-seeded.

Question 99 [Answer: (1)] - Biology (NCERT Page 98)

Explanation: 70S ribosomes are composed of two subunits: 50S (large) and 30S (small).

Question 100 [Answer: (1)] - Biology (NCERT Page 100)

Explanation: Mitochondria and chloroplasts are semi-autonomous organelles containing circular DNA and 70S ribosomes.

Question 101 [Answer: (1)] - Biology (NCERT Page 95, 97-99)

Explanation: Thylakoids: sacs in chloroplast stroma; Cristae: infoldings in mitochondria; Cisternae: disc sacs in Golgi; Chromatin: nucleoprotein fibres.

Question 102 [Answer: (1)] - Biology (NCERT Page 94)

Explanation: Plant cells universally possess a cellulosic cell wall, which is absent in animal cells.

Question 103 [Answer: (1)] - Biology (NCERT Page 108, 109)

Explanation: Lipids have molecular weights < 800 Da and do not form polymers; when tissue is ground, membranes break into vesicles that do not dissolve in acid, retaining them with macromolecules.

Question 104 [Answer: (1)] - Biology (NCERT Page 97)

Explanation: Polar heads face outwards towards the aqueous environment, while hydrophobic tails are directed towards inner part. Thus C is incorrect, A and B are correct.

Question 105 [Answer: (1)] - Biology (NCERT Page 62)

Explanation: Bilateral symmetry in flowers (divided into halves only along one plane) is termed zygomorphic (e.g., Pea, Gulmohur, Bean, Cassia).

Question 106 [Answer: (1)] - Biology (NCERT Page 95)

Explanation: The endomembrane system includes ER, Golgi, lysosomes, and vacuoles. Mitochondria, chloroplasts, and peroxisomes are not part of it.

Question 107 [Answer: (1)] - Biology (NCERT Page 96, 97)

Explanation: The inner membrane of chloroplast is relatively less permeable than the outer membrane. Statements A, B, and C are correct.

Question 108 [Answer: (1)] - Biology (NCERT Page 64)

Explanation: When stamens are attached to the petals, they are epipetalous as in brinjal.

Question 109 [Answer: (1)] - Biology (NCERT Page 110)

Explanation: Amylose in starch forms helical secondary structures that accommodate and trap iodine molecules in the helical core, giving blue color.

Question 110 [Answer: (1)] - Biology (NCERT Page 97)

Explanation: In animal cells, steroidal hormones are synthesized in the Smooth Endoplasmic Reticulum (SER).

Question 111 [Answer: (1)] - Biology (NCERT Page 66)

Explanation: In coconut, the edible part is the cellular and nuclear endosperm. In mango, it is the fleshy mesocarp.

Question 112 [Answer: (1)] - Biology (NCERT Page 98)

Explanation: Svedberg unit (S) stands for sedimentation coefficient; it is indirectly a measure of density and size.

Question 113 [Answer: (1)] - Biology (NCERT Page 89)

Explanation: Both statements are true facts as described in NCERT Anatomy of Flowering Plants.

Question 114 [Answer: (1)] - Biology (NCERT Page 99, 101, 102)

Explanation: Kinetochore: disc structure for spindle attachment; Centromere: primary constriction; Satellite: secondary constriction; Acrocentric: centromere close to end.

Question 115 [Answer: (1)] - Biology (NCERT Page 66, 67)

Explanation: In dicots like gram, pea, and bean, the endosperm is completely consumed during embryo development (non-endospermic). In castor and monocots, endosperm persists.

Question 116 [Answer: (1)] - Biology (NCERT Page 95, 96)

Explanation: Proteins enter RER lumen, are packaged into transport vesicles, fuse with the cis (forming) face of Golgi, pass through cisternae to trans (maturing) face, bud off in secretory vesicles, and fuse with plasma membrane.

Question 117 [Answer: (1)] - Biology (NCERT Page 76)

Explanation: In a dorsiventral leaf, stomata are more frequent on the lower (abaxial) surface, and the mesophyll is distinctly differentiated into palisade and spongy parenchyma.

Question 118 [Answer: (1)] - Biology (NCERT Page 65, 66)

Explanation: In coconut (a drupe), the epicarp is thin, mesocarp is fibrous (yielding coir), and endocarp is stony hard.

Question 119 [Answer: (1)] - Biology (NCERT Page 99)

Explanation: All statements A, B, C, and D accurately describe the 9+2 ultrastructure and origin of eukaryotic cilia and flagella.

Question 120 [Answer: (1)] - Biology (NCERT Page 65)

Explanation: Marginal: Pea; Axile: Tomato/Lemon/China rose; Parietal: Mustard/Argemone; Free central: Dianthus/Primrose.

Question 121 [Answer: (1)] - Biology (NCERT Page 90, 91)

Explanation: Gram-negative bacteria possess lipopolysaccharide in their outer membrane, not Gram-positive. A, B, and C are correct.

Question 122 [Answer: (1)] - Biology (NCERT Page 101)

Explanation: A telocentric chromosome has a terminal centromere located right at the tip.

Question 123 [Answer: (1)] - Biology (NCERT Page 73)

Explanation: A vascular bundle is designated as "closed" specifically because it lacks intrafascicular cambium and hence cannot produce secondary tissues.

Question 124 [Answer: (1)] - Biology (NCERT Page 67)

Explanation: In the grass family, the single cotyledon is shield-shaped and termed the scutellum.

Question 125 [Answer: (1)] - Biology (NCERT Page 64)

Explanation: When stamens are fused into two bundles (e.g., 9 + 1 in pea), it is termed diadelphous.

Question 126 [Answer: (1)] - Biology (NCERT Page 68)

Explanation: Direct elongation of the radicle leads to the formation of the primary tap root.

Question 127 [Answer: (1)] - Biology (NCERT Page 95-98)

Explanation: Golgi: packaging; Lysosomes: intracellular digestion; Mitochondria: cellular respiration/ATP; Ribosomes: protein synthesis.

Question 128 [Answer: (1)] - Biology (NCERT Page 73)

Explanation: Radial: roots; Conjoint open: dicot stem; Conjoint closed: monocot stem; Bicollateral: cucurbits.

Question 129 [Answer: (1)] - Biology (NCERT Page 101)

Explanation: Secondary meristems (like interfascicular cambium and phellogen) originate by dedifferentiation of differentiated permanent cells.

Question 130 [Answer: (1)] - Biology (NCERT Page 74)

Explanation: The endodermis is the innermost cortex layer in dicot roots, with tangential and radial walls impregnated with water-impermeable suberin forming Casparian strips.

Question 131 [Answer: (1)] - Biology (NCERT Page 64)

Explanation: Epipetalous: Brinjal; Epiphyllous: Lily; Monadelphous: China rose; Polyadelphous: Citrus.

Question 132 [Answer: (1)] - Biology (NCERT Page 62)

Explanation: By definition, in a hypogynous flower (mustard, china rose, brinjal), gynoecium sits at top and calyx, corolla, stamens are below it, making ovary superior.

Question 133 [Answer: (1)] - Biology (NCERT Page 95)

Explanation: Lysosomes and vacuoles are single-membrane bound organelles, whereas mitochondria, chloroplasts, and nuclei are double-membrane bound.

Question 134 [Answer: (1)] - Biology (NCERT Page 110)

Explanation: Unlike starch, cellulose does not have helices to hold I₂ molecules, so it gives no color with iodine.

Question 135 [Answer: (1)] - Biology (NCERT Page 76)

Explanation: In roots, protoxylem lies towards periphery and metaxylem towards centre (exarch xylem). In stems, protoxylem lies towards centre (endarch).

Question 136 [Answer: (1)] - Biology (NCERT Page 110, 111)

Explanation: Starch is a polymer of glucose, not galactose. A, B, and C are true facts.

Question 137 [Answer: (1)] - Biology (NCERT Page 106)

Explanation: Uracil is a pyrimidine present exclusively in RNA in place of thymine.

Question 138 [Answer: (1)] - Biology (NCERT Page 112)

Explanation: The first amino acid of a polypeptide has a free amino group and is called the N-terminal amino acid.

Question 139 [Answer: (1)] - Biology (NCERT Page 117)

Explanation: Malonate closely resembles the substrate succinate in structure and competes for the active site of succinate dehydrogenase (competitive inhibition).

Question 140 [Answer: (1)] - Biology (NCERT Page 111, 112)

Explanation: Tertiary structure is essential for the biological catalytic activity of enzymes, not absent or destroyed.

Question 141 [Answer: (1)] - Biology (NCERT Page 106)

Explanation: Nucleoside = Nitrogen base + Sugar. Nucleotide = Nitrogen base + Sugar + Phosphate group.

Question 142 [Answer: (1)] - Biology (NCERT Page 106)

Explanation: Palmitic acid: 16 C; Arachidonic: 20 C; Glycerol: trihydroxy propane; Lecithin: membrane phospholipid.

Question 143 [Answer: (1)] - Biology (NCERT Page 111)

Explanation: All statements A, B, C, and D are classical Watson-Crick B-DNA and nucleic acid biochemistry facts from NCERT.

Question 144 [Answer: (1)] - Biology (NCERT Page 106)

Explanation: Adenosine: nucleoside; Adenylic acid: nucleotide; Adenine: purine base; DNA: polynucleotide.

Question 145 [Answer: (1)] - Biology (NCERT Page 109)

Explanation: As per NCERT Table 9.4: Water: 70-90%, Proteins: 10-15%, Nucleic acids: 5-7%, Carbohydrates: 3%, Lipids: 2%, Ions: 1%.

Question 146 [Answer: (1)] - Biology (NCERT Page 112)

Explanation: Primary: linear sequence; Secondary: $\pm$ -helix/ 2 -sheet; Tertiary: 3D conformation; Quaternary: spatial arrangement.

Question 147 [Answer: (1)] - Biology (NCERT Page 112)

Explanation: Enzymes accelerate reaction velocity by lowering the activation energy barrier between ground state and transition state complex.

Question 148 [Answer: (1)] - Biology (NCERT Page 108)

Explanation: NCERT Table 9.3: Alkaloids: Morphine/Codeine; Toxins: Abrin/Ricin; Lectins: Concanavalin A; Drugs: Vinblastin/Curcumin.

Question 149 [Answer: (1)] - Biology (NCERT Page 110, 111)

Explanation: Cellulose: linear 2 -glucan; Inulin: fructan; Glycogen: branched animal glucan; Starch: helical plant glucan.

Question 150 [Answer: (1)] - Biology (NCERT Page 106, 111)

Explanation: Ribulose biphosphate carboxylase-oxygenase (RuBisCO) is the most abundant protein in the entire biosphere. Collagen is the most abundant in the animal world.

Question 151 [Answer: (1)] - Biology (NCERT Page 106, 108)

Explanation: Glutamic acid is acidic; Lysine is basic; Valine is neutral; Tyrosine, Tryptophan, and Phenylalanine are aromatic.

Question 152 [Answer: (1)] - Biology (NCERT Page 108)

Explanation: Curcumin is classified as a Drug, not a monoterpene (Monoterpenes include pinene, camphor, etc.).

Question 153 [Answer: (1)] - Biology (NCERT Page 106)

Explanation: Standard IUBMB 6 classes: Oxidoreductases, Transferases, Hydrolases, Lyases, Isomerases, Ligases.

Question 154 [Answer: (1)] - Biology (NCERT Page 109)

Explanation: As per NCERT Table 9.1, Oxygen comprises 65.0% of the human body by weight, followed by Carbon (18.5%).

Question 155 [Answer: (1)] - Biology (NCERT Page 112, 113)

Explanation: Enzymes function over a narrow range and display maximum velocity at optimum temperature and optimum pH.

Question 156 [Answer: (1)] - Biology (NCERT Page 110)

Explanation: Glycogen has $\pm$ -1,4-glucosidic bonds in linear chains and $\pm$ -1,6-glucosidic bonds at branching points.

Question 157 [Answer: (1)] - Biology (NCERT Page 110, 111)

Explanation: Humans lack the enzyme cellulase and cannot digest cellulose.

Question 158 [Answer: (1)] - Biology (NCERT Page 108)

Explanation: Collagen forms fibrous structural networks throughout extracellular matrix in connective tissues across animals.

Question 159 [Answer: (1)] - Biology (NCERT Page 114, 118)

Explanation: Coenzymes are loosely and transiently bound, and catalytic activity is lost if cofactors are removed, but they do not always remain irreversibly bound.

Question 160 [Answer: (1)] - Biology (NCERT Page 108)

Explanation: Haem: prosthetic group of catalase/peroxidase; Zn^{2+} : carboxypeptidase; NAD: contains niacin; Mg^{2+} : hexokinase.

Question 161 [Answer: (1)] - Biology (NCERT Page 117)

Explanation: Official IUBMB definitions as detailed in NCERT Biomolecules chapter.

Question 162 [Answer: (1)] - Biology (NCERT Page 108, 109)

Explanation: Both statements are directly taken from NCERT text under How to Analyse Chemical Composition.

Question 163 [Answer: (1)] - Biology (NCERT Page 115)

Explanation: Enzymes lower the activation energy barrier without altering the initial or final energy states or "G."

Question 164 [Answer: (1)] - Biology (NCERT Page 117)

Explanation: Apoenzyme (protein) + Cofactor = Holoenzyme (active complex).

Question 165 [Answer: (1)] - Biology (NCERT Page 109-111)

Explanation: Proteins: amino acids; Nucleic acid: nucleotides; Cellulose: glucose; Inulin: fructose.

Question 166 [Answer: (1)] - Biology (NCERT Page 110)

Explanation: Heartwood (duramen): central dark durable dead wood; Sapwood (alburnum): light conducting wood; Phellem: outer cork; Phelloderm: inner secondary cortex.

Question 167 [Answer: (1)] - Biology (NCERT Page 111, 112)

Explanation: Hemoglobin is an exemplary multimeric protein with 2 α and 2 β subunits arranged in quaternary conformation.

Question 168 [Answer: (1)] - Biology (NCERT Page 118)

Explanation: By definition, secondary metabolites do not have identifiable direct roles in primary growth, development, or respiration in the host organism.

Question 169 [Answer: (1)] - Biology (NCERT Page 118)

Explanation: Haem is the prosthetic group for both catalase and peroxidase enzymes.

Question 170 [Answer: (1)] - Biology (NCERT Page 105, 106)

Explanation: Palmitic acid has 16 carbons, whereas arachidonic acid has 20 carbons. A, C, and D are correct.

Question 171 [Answer: (1)] - Biology (NCERT Page 110)

Explanation: Both starch and glycogen are homopolymers composed entirely of α -D-glucose units.

Question 172 [Answer: (1)] - Biology (NCERT Page 115)

Explanation: Classical sequence of catalytic cycle as detailed in NCERT.

Question 173 [Answer: (1)] - Biology (NCERT Page 110, 111)

Explanation: Starch reacts with iodine to give an intense blue-black color, not yellow. A and B are correct.

Question 174 [Answer: (1)] - Biology (NCERT Page 117, 118)

Explanation: Coenzymes NAD and NADP contain the vitamin niacin.

Question 175 [Answer: (1)] - Biology (NCERT Page 106)

Explanation: NCERT opening sentence of enzymes section: "Almost all enzymes are proteins. There are some nucleic acids that behave like enzymes. These are called ribozymes."

Question 176 [Answer: (1)] - Biology (NCERT Page 111)

Explanation: Both statements are scientifically accurate and conform directly to NCERT Biomolecules.

Question 177 [Answer: (1)] - Biology (NCERT Page 110, 112)

Explanation: A competitive inhibitor increases the apparent K_m , but does NOT alter the V_{max} (since sufficiently high substrate concentration overcomes the competition).

Question 178 [Answer: (1)] - Biology (NCERT Page 116)

Explanation: K_m (Michaelis constant) is defined as the substrate concentration at which the velocity of the reaction is equal to half of its maximum velocity ($V_{max}/2$).

Question 179 [Answer: (1)] - Biology (NCERT Page 118)

Explanation: Prosthetic: tightly bound organic; Coenzyme: transient organic; Metal ion: inorganic coordination; Apoenzyme: protein.

Question 180 [Answer: (1)] - Biology (NCERT Page 105, 109)

Explanation: NCERT Table 9.1: Hydrogen: 0.5%, Carbon: 18.5%, Nitrogen: 3.3%, Oxygen: 65.0%.